

Complex Numbers

- **Imaginary Unit:** The imaginary unit is defined as:

$$i = \sqrt{-1}, \quad \text{so that } i^2 = -1$$

- **Complex Number:** A complex number is any number of the form:

$$z = a + bi$$

where:

- a is the real part (a real number).
- bi is the imaginary part (a real number multiplied by i).

- **Pure Imaginary Number:** If $a = 0$, then the complex number $z = bi$ is called a *pure imaginary number*.

- **Addition of Complex Numbers:**

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

- **Multiplication of Complex Numbers:**

$$(a + bi) \cdot (c + di) = (ac - bd) + (ad + bc)i$$

- **Conjugate of a Complex Number:** The conjugate of a complex number $z = a + bi$ is denoted as $\bar{z}$ and is defined as:

$$\bar{z} = a - bi$$

- **Modulus (Magnitude) of a Complex Number:** The modulus or magnitude of a complex number $z = a + bi$ is given by:

$$|z| = \sqrt{a^2 + b^2}$$

- **Division of Complex Numbers:**

$$\frac{a + bi}{c + di} = \frac{(a + bi) \cdot (c - di)}{(c + di) \cdot (c - di)} = \frac{(a + bi) \cdot (c - di)}{c^2 + d^2}$$